

Portfolio Analysis and Optimisation of a Diversified U.S. Equity Portfolio

An Application of Modern Portfolio Theory

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Abstract

This report is a quantitative analysis of a diversified portfolio of 30 U.S. equities across seven sectors: technology, financials, healthcare, consumer goods, automobiles, energy and industrials. It covers the five-year period from January 2020 to January 2025. Using historical closing price data from Yahoo Finance, daily and cumulative returns were calculated for each stock. An equally weighted portfolio was then built, with each stock given a weight of 3.33%, and evaluated using standard Modern Portfolio Theory techniques.

The equal-weighted portfolio delivered an expected annual return of 23.78%, annualised volatility of 21.40%, and a Sharpe Ratio of 0.92. It also suffered a maximum drawdown of 33.24% during the COVID-19 market shock of early 2020. To approximate the Efficient Frontier, a Monte Carlo simulation of 10,000 randomly weighted portfolios was run. This identified a Maximum Sharpe Portfolio (29.53% return, 22.60% volatility, Sharpe Ratio of 1.13) and a Minimum Volatility Portfolio (20.44% return, 18.55% volatility, Sharpe Ratio of 0.89).

These results show that how capital is allocated across a portfolio, not just which assets are chosen, is a major driver of risk-adjusted performance. Shifting weight toward Apple, Exxon Mobil, Costco, Meta Platforms and Amazon made the portfolio more efficient than the equal-weight benchmark, without taking on proportionally more risk. The report closes with a discussion of what these findings mean for portfolio construction in practice, and where Modern Portfolio Theory falls short in real markets.

Keywords: Modern Portfolio Theory, portfolio optimisation, Monte Carlo simulation, Efficient Frontier, Sharpe Ratio, diversification, risk-adjusted return

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1. Introduction

1.1 Background

Anyone building a portfolio of risky assets has to weigh expected return against risk. Modern Portfolio Theory, developed by Harry Markowitz, gives investors a framework for measuring and optimising that trade-off, and it still underpins most portfolio construction methods used today. This report applies that framework to a diversified portfolio of U.S. equities, using a Python-based workflow that runs from data collection through to portfolio optimisation.

All the calculations, figures and outputs behind this report come from an original Jupyter notebook. This report simply restructures and explains that work for academic and professional readers. As a next step, the outputs have also been set up for an interactive Streamlit dashboard, so a non-technical audience can explore the portfolio's performance, risk and optimisation results for themselves.

1.2 Scope of the Study

This study looks only at a portfolio of 30 U.S.-listed equities across seven sectors, using daily closing prices from 1 January 2020 to 1 January 2025. It starts with an equally weighted allocation as a neutral baseline, then moves on to portfolio optimisation through Monte Carlo simulation. It does not cover transaction costs, taxes, rebalancing over time, short selling, out-of-sample testing, or asset classes outside public equities. Every result comes from historical data, so it describes past performance rather than predicting future returns.

2. Objectives

2.1 Research Aim

This study aims to build, evaluate and optimise a diversified portfolio of U.S. equities using the principles of Modern Portfolio Theory, and to present the analysis in a way suited to both academic and professional readers.

2.2 Research Objectives

To reach that aim, the study set out to:

- Download and validate historical stock price data for a diversified set of U.S. equities.
- Calculate daily and cumulative returns for each constituent stock.
- Construct an equally weighted portfolio and compute its expected return and volatility.
- Analyse the covariance and correlation between the returns of the constituent assets.
- Evaluate portfolio performance using risk-adjusted measures, including the Sharpe Ratio, rolling volatility and maximum drawdown.

- Generate a large number of randomly weighted portfolios through Monte Carlo simulation to approximate the Efficient Frontier.
- Identify and interpret the Maximum Sharpe and Minimum Volatility portfolios, and communicate the findings in a structured research report.

3. Literature Review

Modern Portfolio Theory is built on a simple idea: a rational investor should aim to get the highest possible return for a given level of risk, or the lowest possible risk for a given level of return, rather than judging assets one by one. Its key contribution is to put diversification on a formal footing. Because individual assets don't move in perfect lockstep with each other, combining them in a portfolio can lower overall risk without giving up a proportional amount of return. This works through covariance and correlation, which measure how far two assets move together and how strong and in what direction that relationship is, on a scale from minus one to plus one. Portfolios built from weakly or negatively correlated assets gain more from diversification than portfolios built from highly correlated ones, because offsetting price movements smooth out overall volatility.

A portfolio's risk-adjusted performance is usually measured with the Sharpe Ratio: the return earned above the risk-free rate, divided by the volatility taken on to earn it. A higher Sharpe Ratio means more return for each unit of risk, which is why the measure is so widely used to compare portfolios with different risk profiles. Building on this idea, the Efficient Frontier is the set of portfolios that offer the highest expected return for each level of risk. Any portfolio sitting below the frontier is inefficient, since a better alternative exists, either a higher return for the same risk or the same return for less risk. Two points on the frontier matter most here: the Maximum Sharpe Portfolio, which gives the best risk-adjusted return, and the Minimum Volatility Portfolio, which gives the lowest risk achievable.

For a large number of assets, working out the Efficient Frontier exactly means solving a constrained quadratic optimisation problem, which is not straightforward. In practice it's more common to approximate the frontier using Monte Carlo simulation: generate a large number of randomly weighted portfolios, calculate each one's expected return, volatility and Sharpe Ratio, then use the resulting spread of points to trace out the boundary of efficient allocations. This is the approach used throughout Section 6.4 of this report, and it's a practical, widely used way to approximate the frontier without needing an exact mathematical solution.

A well-known weakness of the mean-variance framework, whether applied analytically or through simulation, is that its inputs, expected returns, variances and covariances, all have to be estimated from historical data. Expected returns in particular are notoriously harder to estimate accurately than volatility or correlation. Because mean-variance optimisation is highly sensitive to the return assumptions fed into it, even small estimation errors can produce large swings in the resulting optimal allocation. This is the reason extensions like the Black-Litterman model exist, discussed further in Section 7.6, and it's worth keeping in mind when interpreting the optimisation results in this report.

For all its theoretical elegance, Modern Portfolio Theory rests on assumptions that don't hold exactly in real markets: that returns are normally distributed, that correlations stay stable over time, and that investors can trade without cost. Section 7 looks at these limitations more closely, and what they mean for this particular analysis.

4. Methodology

This section applies the concepts from Section 3 to the portfolio described in Section 5. Expected portfolio return was calculated as the dot product of the weight vector and the vector of mean daily returns for each stock, then annualised by multiplying by 252, the standard number of trading days in a year:

$$E[Rp] = (w^T \cdot \mu) \times 252$$

Here w is the vector of portfolio weights and μ is the vector of mean daily returns. Portfolio volatility was calculated as the square root of the weight vector multiplied by the covariance matrix of daily returns and its transpose, again annualised, this time using the square root of 252:

$$\sigma_p = \sqrt{(w^T \Sigma w)} \times \sqrt{252}$$

Here Σ is the covariance matrix of daily returns. This formula captures not just each asset's own volatility but how far the assets move together, which the covariance matrix picks up. The Sharpe Ratio was then calculated as:

$$Sharpe = (E[Rp] - R_f) / \sigma_p$$

Here R_f is the assumed annual risk-free rate of 4%, roughly the yield on short-term U.S. Treasuries. Rolling volatility was calculated as the annualised standard deviation of portfolio returns over a moving 30-day window. A 30-day window was chosen because it reacts quickly to short-term shifts in the market while still being smooth enough to pick out sustained periods of higher risk. Maximum drawdown was calculated as:

$$Drawdown(t) = (Cumulative(t) - RunningMax(t)) / RunningMax(t)$$

This tracks the portfolio's running maximum value over time and finds the largest percentage fall from that peak at any point in the sample period.

The Efficient Frontier was approximated by running a Monte Carlo simulation of 10,000 randomly weighted portfolios, with weights always summing to 100% and no short selling allowed. For each simulated portfolio, expected annual return, annualised volatility and Sharpe Ratio were worked out using the formulas above. A fixed random seed (`np.random.seed(18)`) was used so the simulation can be reproduced. The portfolio with the highest Sharpe Ratio in the simulated set became the Maximum Sharpe Portfolio, and the one with the lowest volatility became the Minimum Volatility Portfolio.

5. Data Collection

5.1 Data Source

Historical daily price data were pulled using the yfinance Python library, which gives programmatic access to Yahoo Finance, covering the five years from 1 January 2020 to 1 January 2025. Prices were downloaded on an adjusted-close basis (auto_adjust=True), so dividends and stock splits are accounted for and the return calculations reflect total investment performance, not just price movement.

5.2 Portfolio Selection and Composition

The portfolio was built to span seven major sectors of the U.S. economy rather than concentrate on one industry, which cuts concentration risk and spreads exposure across different economic drivers such as interest rates, consumer spending, commodity prices and business investment. Technology was given the largest number of stocks, reflecting how much weight it now carries in U.S. equity markets.

Table 5.1: Portfolio Constituents by Sector

Sector	Constituent Tickers	No. of Stocks
Technology	AAPL, MSFT, NVDA, AMZN, GOOG, META, AMD, ORCL	8
Financials	JPM, BAC, GS, V, MA	5
Healthcare	JNJ, PFE, LLY, MRK, ABBV	5
Consumer	WMT, COST, PG, KO, MCD	5
Automobiles	TSLA, GM, F	3
Energy	XOM, CVX	2
Industrials	CAT, GE	2

This adds up to 30 equities in total. Amazon and Alphabet are grouped under Technology here, matching how they were classified in the notebook, since both companies have significant technology and cloud computing operations even though standard industry taxonomies would formally place them under consumer discretionary or communication services.

5.3 Data Preprocessing and Portfolio Construction

After downloading, the dataset was checked for structure, dimensions and missing values. Daily percentage returns were then calculated for each stock using pct_change(), and the first row, which is always undefined, was dropped using dropna(). Working with returns rather than raw prices puts every stock on the same scale regardless of its price, and this is the foundation for every calculation that follows: cumulative returns, covariance, correlation, portfolio-level return and volatility, the Sharpe Ratio, and the Monte Carlo simulation.

The baseline portfolio used equal weighting, with each of the 30 stocks given an identical share of 1/30, or approximately 3.33%. The weights DataFrame (weights_df) confirmed these summed to 100%. Equal weighting was chosen as the starting point because it's a neutral benchmark that doesn't assume anything about which stocks are more attractive than others, so any differences in portfolio performance come purely from the stocks' own returns and how they interact with each other.

6. Results

6.1 Correlation and Covariance Analysis

The covariance and correlation matrices of daily returns were computed across all 30 constituents. Printing the full 30-by-30 matrix would add little on the page, so it isn't reproduced here. The complete matrix sits in the notebook alongside the rest of the workings.

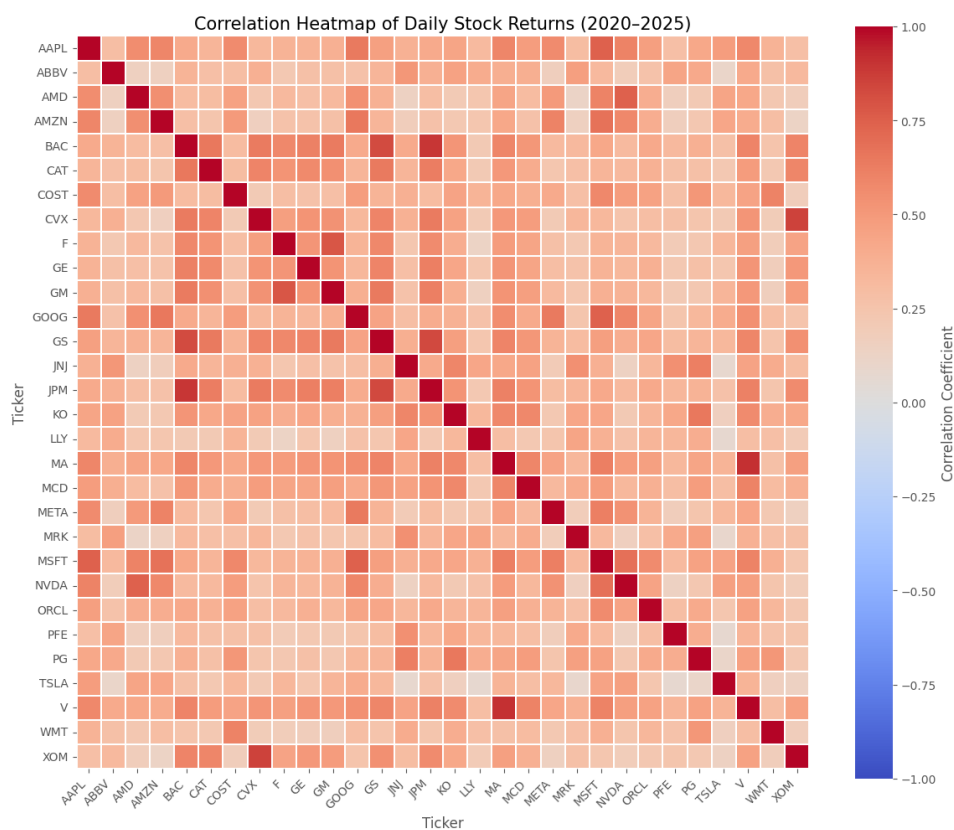


Figure 6.1: Correlation Heatmap of Daily Stock Returns (2020-2025)

The technology names, Apple, Microsoft, NVIDIA, Alphabet, Amazon and Advanced Micro Devices, show fairly strong positive correlation with each other, which makes sense given they share the same innovation-driven growth story. The financial stocks, JPMorgan, Bank of America and Goldman Sachs, are similarly correlated, since they're all sensitive to interest rates and the broader economy, and the same

goes for Chevron and Exxon Mobil, which both move with energy prices. Healthcare and consumer staples names, such as Johnson & Johnson, Pfizer, Procter & Gamble and Walmart, correlate more weakly with the technology cluster. It's this variation across sectors, rather than any negatively correlated pairs (there aren't any in this dataset), that gives the portfolio most of its diversification benefit.

6.2 Return and Growth Analysis

Looking at daily returns, NVIDIA (NVDA) had the highest average at 0.3050%, followed by Tesla (TSLA) at 0.2997% and Eli Lilly (LLY) at 0.1663%. Pfizer (PFE) had the lowest average daily return at just 0.0052%, with Johnson & Johnson and Coca-Cola not far behind it.

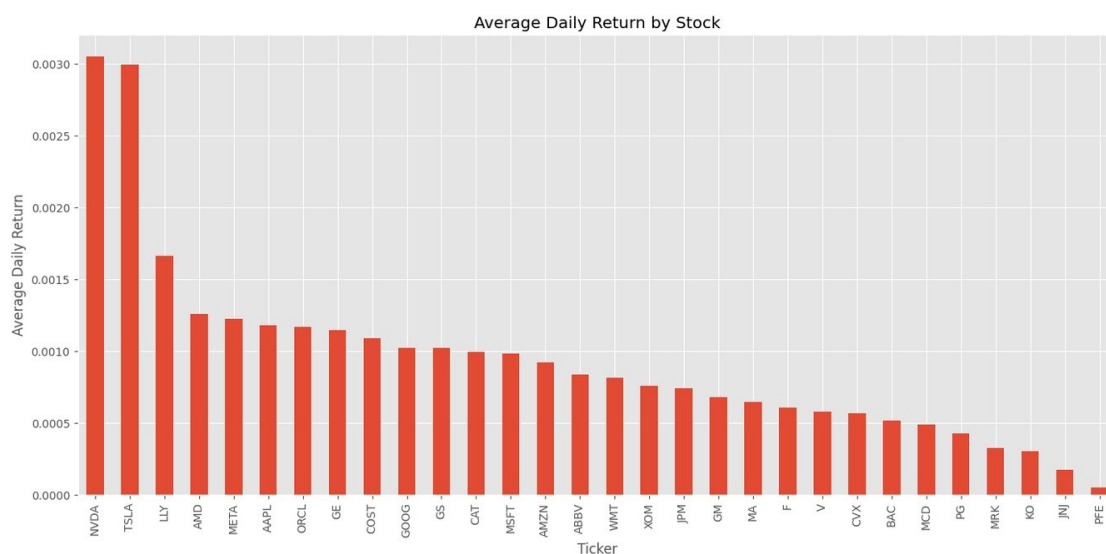


Figure 6.2: Average Daily Return by Stock

Returns are spread unevenly across the portfolio. Growth-oriented technology and healthcare names produced average daily returns well above the more defensive stocks, a pattern that hints at the risk-return trade-offs covered in Section 6.3.

Cumulative returns, the running product of one plus the daily return series, show that the five strongest performers over the long run were NVIDIA, Tesla, Eli Lilly, Apple and Costco.

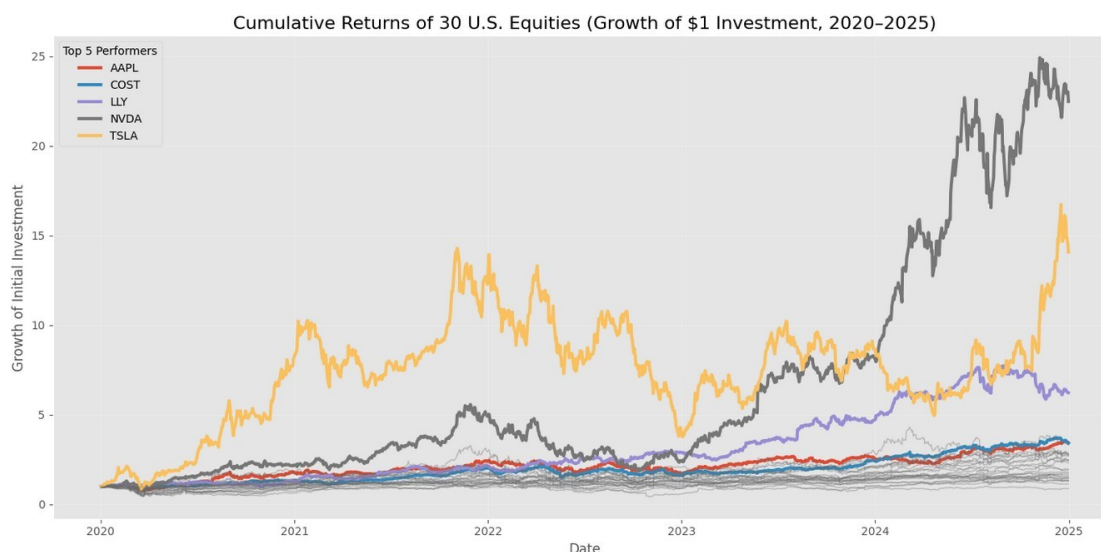


Figure 6.3: Cumulative Returns of 30 U.S. Equities (Growth of a \$1 Investment, 2020-2025)

NVIDIA came out well ahead on cumulative return despite some sharp swings between 2022 and 2023, while Tesla had the bumpiest ride of the top performers but still delivered exceptional long-term growth. Eli Lilly, Apple and Costco climbed more steadily, which shows that the highest average daily return doesn't automatically mean the highest cumulative return, since compounding depends on the order returns arrive in, not just their average. Overall, the biggest long-term gains tended to come with more volatility, while the steadier, lower-volatility growth of Eli Lilly, Apple and Costco produced stronger risk-adjusted results.

This gap between the return rankings in Figure 6.2 and the growth rankings in Figure 6.3 comes down to a well-known relationship between arithmetic and geometric returns: the compounded return an investor actually earns is always less than or equal to the simple average of the periodic returns, and that gap widens as volatility rises, an effect known as volatility drag. A stock with a high average daily return but a lot of day-to-day swing, like Tesla, ends up giving back more of its arithmetic performance to compounding than a stock with a lower but steadier average, like Costco. This is why the stocks ranked by average daily return in Section 6.2 don't line up exactly with the stocks ranked by cumulative growth here, and it's a reminder that volatility is a real cost to a long-term investor, not just an abstract risk number.

6.3 Risk Analysis

The equal-weighted portfolio had a daily volatility of about 1.35%, which annualises to 21.40%, against an expected annual return of 23.78% (Section 6.2). This level of risk reflects the portfolio's exposure to higher-volatility growth names like NVIDIA, Tesla and Meta, partly offset by the diversification benefits from Section 6.1, which kept overall portfolio risk well below the level of its most volatile individual holdings.

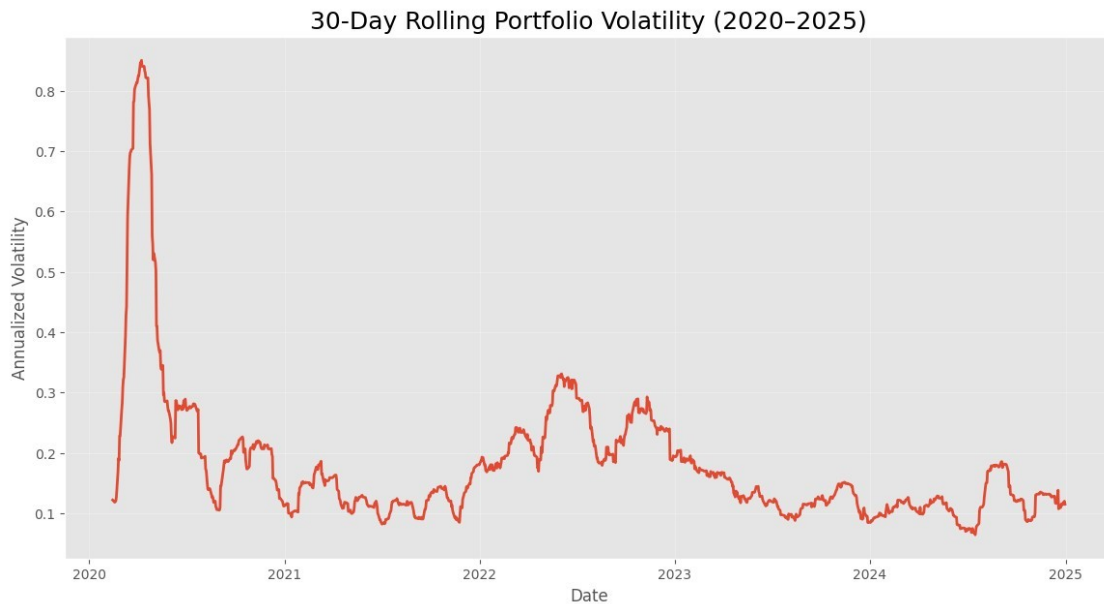


Figure 6.4: 30-Day Rolling Portfolio Volatility (2020-2025)

Rolling volatility averaged 18.10% over the sample period, but it spiked to 85.03% in the early months of 2020, right as COVID-19 hit, before dropping to a low of 6.42% once markets settled down. A second check using a 90-day window showed the same spike, just smoother, confirming that the elevated risk was a genuine market event and not just an artefact of the 30-day window chosen. This shows that portfolio risk moves over time rather than staying fixed: it climbs sharply under acute market stress and eases off again once conditions normalise.

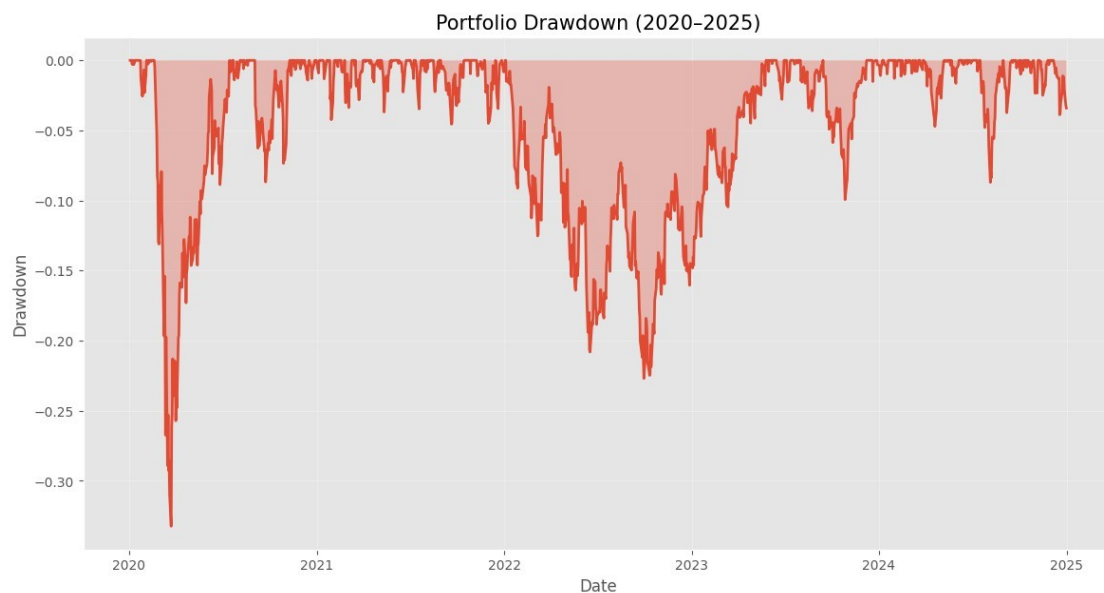


Figure 6.5: Portfolio Drawdown (2020-2025)

The portfolio's maximum drawdown was 33.24%, meaning an investor who bought at the previous peak would have watched their holding fall by roughly a third before it recovered. Given that the sample period includes the COVID-19 crash, the 2022 bear market and a stretch of rapidly rising interest rates, a drawdown of this size is about what you'd expect from a portfolio with a heavy tilt toward growth technology stocks. The drawdown series also shows the portfolio repeatedly climbing back to new highs, which points to a fair degree of resilience despite these rough patches.

How much a 33.24% drawdown actually matters depends heavily on an investor's time horizon and need for cash. Someone who stayed invested through the whole decline would, based on the recovery seen in the drawdown series, have made the loss back within the five-year sample. Someone forced to sell part of the portfolio during the trough, whether because of a fixed withdrawal schedule or a margin call, would end up with a much worse outcome than the annualised return and volatility figures alone suggest. This is what's known as sequence-of-returns risk. It's the timing of cash flows relative to a drawdown, not the average return over the full period, that decides the real outcome for anyone who isn't purely buy-and-hold. That makes maximum drawdown a genuinely different risk measure from volatility. Volatility describes the average spread of returns. Drawdown captures the worst-case outcome for an investor unlucky enough to need cash at the worst possible moment.

Using an assumed annual risk-free rate of 4%, the equal-weighted portfolio achieved a Sharpe Ratio of 0.924, meaning it earned about 0.92 units of excess return for every unit of risk taken. That's just under the commonly cited benchmark of 1.0 for strong risk-adjusted performance. In other words, the equal-weight allocation paid investors reasonably well for the risk they took, but it wasn't the most efficient use of capital available, a claim tested directly by the optimisation work in Section 6.4.

6.4 Portfolio Optimisation

A Monte Carlo simulation of 10,000 randomly weighted portfolios was used to approximate the Efficient Frontier and to identify the Maximum Sharpe and Minimum Volatility portfolios described in Section 3.

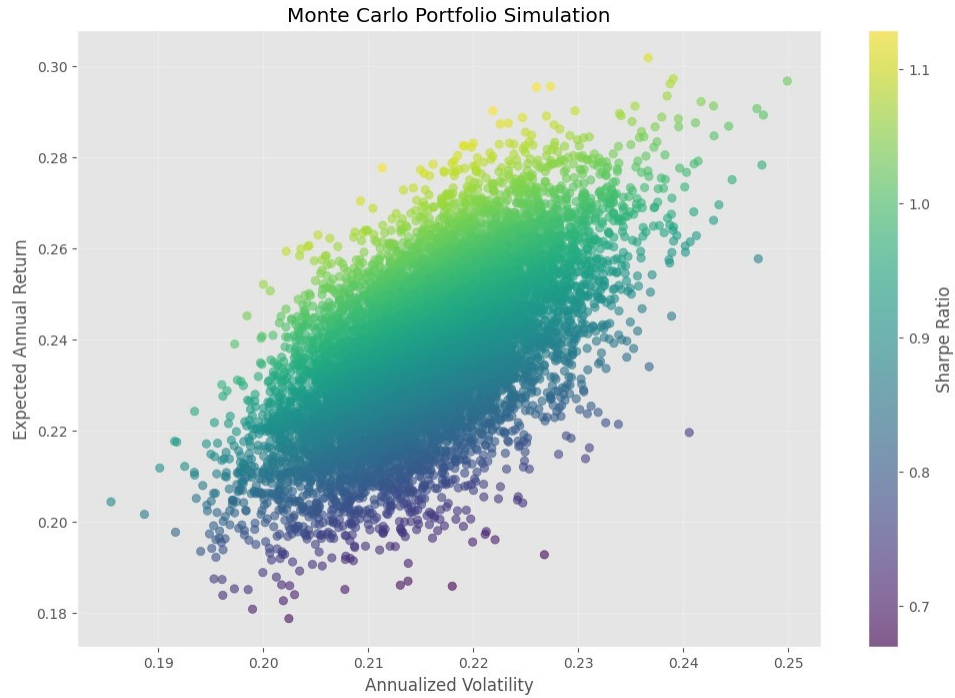


Figure 6.6: Monte Carlo Portfolio Simulation (10,000 Portfolios)

The simulated portfolios trace a clear positive relationship between risk and return, exactly what Modern Portfolio Theory predicts: higher expected returns come with higher volatility. Among the 10,000 simulations, the portfolio with the best Sharpe Ratio achieved an expected annual return of 29.53% at an annualised volatility of 22.60%, giving a Sharpe Ratio of 1.13, compared with 0.92 for the equal-weight benchmark.

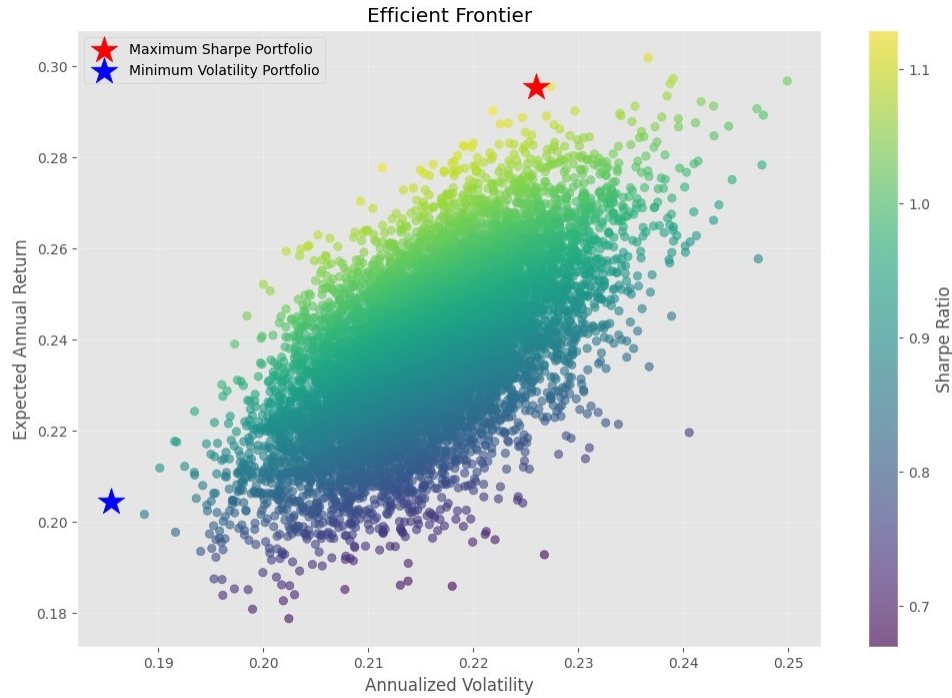


Figure 6.7: Efficient Frontier with Maximum Sharpe and Minimum Volatility Portfolios

Table 6.1: Optimised Portfolios Compared with the Equal-Weight Benchmark

Portfolio	Expected Return	Annualised Volatility	Sharpe Ratio
Equal-Weight Portfolio	23.78%	21.40%	0.92
Maximum Sharpe Portfolio	29.53%	22.60%	1.13
Minimum Volatility Portfolio	20.44%	18.55%	0.89

The Maximum Sharpe Portfolio raises expected return by 5.75 percentage points over the equal-weight benchmark while volatility goes up by only 1.20 percentage points, which tells us the gain comes mainly from allocating capital more efficiently, not from simply taking on more risk. The Minimum Volatility Portfolio, on the other hand, cuts volatility by 2.85 percentage points but gives up 3.34 percentage points of expected return in exchange, and its Sharpe Ratio of 0.89 ends up marginally below the equal-weight benchmark. So minimising risk on its own doesn't automatically improve risk-adjusted efficiency. Section 7 looks at what this trade-off means in practice.

6.5 Portfolio Allocation

The Maximum Sharpe Portfolio does not distribute capital equally. Larger weights were assigned to assets that contributed more favourably to the portfolio's risk-adjusted return.

Table 6.2: Largest Allocations in the Maximum Sharpe Portfolio

Rank	Ticker	Company	Portfolio Weight
1	AAPL	Apple Inc.	7.33%
2	XOM	Exxon Mobil Corporation	7.26%
3	COST	Costco Wholesale Corporation	7.13%
4	META	Meta Platforms, Inc.	7.11%
5	AMZN	Amazon.com, Inc.	6.67%

These five holdings collectively account for approximately 35.5% of total capital. The weights for the remaining five names in the top ten can be read straight off Figure 6.8. The full ranked list across all 30 constituents is available in the notebook.

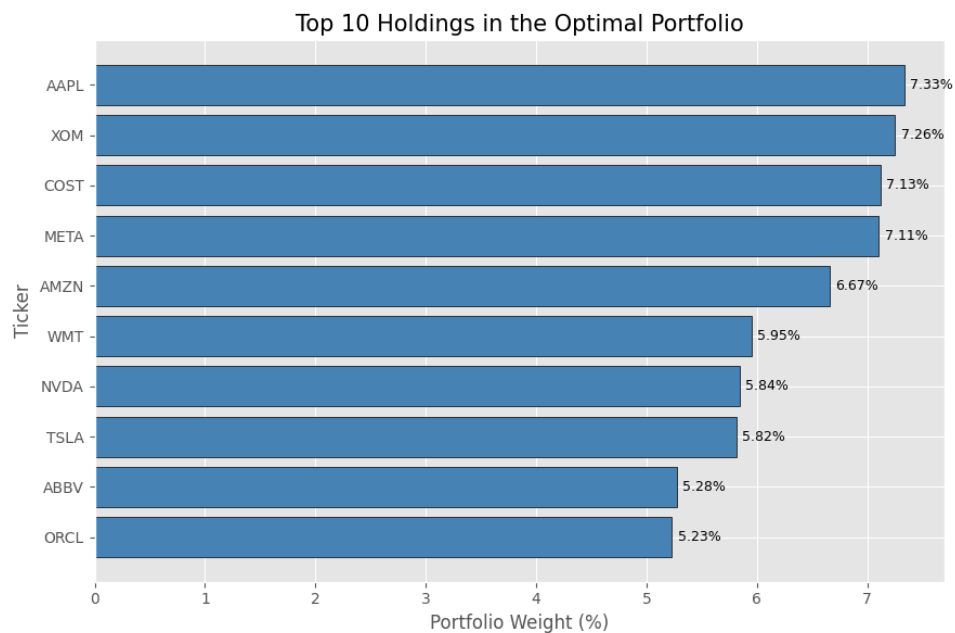


Figure 6.8: Top 10 Holdings in the Optimal Portfolio

The largest positions span Technology (Apple, Meta, Amazon), Energy (Exxon Mobil) and Consumer (Costco) rather than sitting in one sector, and no single stock dominates the portfolio. This spread across sectors shows that the improved Sharpe Ratio from Section 6.4 comes from weighting the existing diversification more efficiently, in line with the correlation patterns from Section 6.1, rather than from piling into a narrow group of correlated stocks. Exxon Mobil's place among the top holdings is worth noting, since it had one of the more modest average daily returns of the 30 stocks. That shows the optimiser rewards a stock for its diversification value as well as its own return: Exxon Mobil's weak

correlation with the technology-heavy core of the portfolio helps cut overall volatility in a way that ranking stocks by return alone would have missed.

6.6 Portfolio Improvement Comparison

Table 6.3 brings together the performance of all three portfolios covered in this study, the original equal-weighted allocation and the two portfolios found through Monte Carlo optimisation, to summarise the overall improvement gained from reallocating capital.

Table 6.3: Summary Comparison of Portfolio Strategies

Portfolio	Expected Return	Annualised Volatility	Sharpe Ratio	Change in Sharpe Ratio vs Equal-Weight
Equal-Weight Portfolio	23.78%	21.40%	0.92	(benchmark)
Maximum Sharpe Portfolio	29.53%	22.60%	1.13	+0.21 (+22.8%)
Minimum Volatility Portfolio	20.44%	18.55%	0.89	-0.03 (-3.3%)

This comparison shows that optimisation improved risk-adjusted performance meaningfully, though not in the same way for every objective. The Maximum Sharpe Portfolio lifted the Sharpe Ratio by about 23% over the equal-weighted benchmark, purely by reallocating toward Apple, Exxon Mobil, Costco, Meta Platforms and Amazon, without changing the underlying 30-stock universe at all. That confirms it was capital allocation, not stock selection, driving the improvement. The Minimum Volatility Portfolio did what it set out to do, cutting annualised volatility by 2.85 percentage points, but its Sharpe Ratio ended up marginally below the equal-weight benchmark, showing that minimising volatility on its own isn't the same as maximising risk-adjusted return. Together, these results back up the central claim of this study: a well-weighted portfolio can beat a naively diversified one on a risk-return basis without changing which assets it holds, though which objective you choose, maximising Sharpe or minimising volatility, shapes the resulting allocation and how well it suits different investors.

7. Discussion

7.1 Portfolio Performance and Optimisation in Context

The main finding here is simple: shifting capital from an equal-weighted position toward Apple, Exxon Mobil, Costco, Meta Platforms and Amazon lifted the portfolio's Sharpe Ratio from 0.92 to 1.13. That backs up one of the core ideas in Modern Portfolio Theory. Risk-adjusted performance depends heavily on how capital is spread across assets, not just on which assets you hold. The equal-weight portfolio and the Maximum Sharpe Portfolio hold exactly the same 30 stocks, so the entire improvement comes from reweighting, not from picking different stocks. That distinction matters in practice. Naive diversification,

spreading money evenly across a chosen set of assets, is often used as a default when there is no more considered allocation strategy in place.

It's worth noting that the optimisation process didn't just pile capital into the portfolio's best historical performers. NVIDIA and Tesla, the two strongest individual stocks from Section 6.2, don't appear among the five largest holdings of the Maximum Sharpe Portfolio in Section 6.5. Instead, capital went toward Apple, Exxon Mobil, Costco, Meta Platforms and Amazon, stocks with solid returns but more moderate volatility, and in Exxon Mobil's case, fairly weak correlation with the technology-heavy core of the portfolio. This is an easy point to miss when you look at return rankings on their own. How much a stock contributes to portfolio efficiency depends jointly on its own return and volatility and on how it moves relative to every other holding, not on its standalone performance.

The jump from 0.92 to 1.13 in Sharpe Ratio is a meaningful improvement, not a marginal one. The modest rise in volatility that came with it, covered in Section 6.4, compounds over a multi-year holding period under normal reinvestment assumptions. So a gain that looks small on an annual basis ends up making a real difference to the investment outcome over the five-year study horizon and beyond. This confirms that the reweighting exercise, rather than the choice of the underlying 30-stock universe, was the main driver of the improvement in risk-adjusted performance.

7.2 The Efficient Frontier and the Risk-Return Trade-off

Comparing the Maximum Sharpe and Minimum Volatility portfolios from Section 6.4 shows that minimising risk and maximising risk-adjusted return are not the same goal. An investor chasing the lowest possible volatility would have accepted a slightly lower Sharpe Ratio (0.89) than the equal-weight benchmark (0.92), in exchange for cutting volatility by 2.85 percentage points. That tells us the Minimum Volatility Portfolio gives up more expected return than it saves in risk. This comes down to the shape of the Efficient Frontier. Portfolios near its lower-risk end tend to have a flatter return profile, so further cuts to volatility cost more in lost return. Portfolios closer to the Maximum Sharpe point sit where extra return can still be picked up relatively cheaply in risk terms.

This has a practical takeaway for how to use the frontier when building a portfolio. Rather than treating the Maximum Sharpe Portfolio as the universally correct choice, an investor should find the point on the frontier that fits their own appetite for risk. A pension fund nearing its payout phase, for example, has a clear capital-preservation goal, so the Minimum Volatility Portfolio's smaller drawdowns might justify its lower Sharpe Ratio even though it's technically less efficient. An investor with a longer horizon and more tolerance for short-term swings would likely prefer the stronger return-to-risk balance offered by the Maximum Sharpe Portfolio. The Efficient Frontier isn't really about finding one correct portfolio. It's a map of the full set of efficient choices, and the right one depends on the investor picking it.

7.3 Significance of the Correlation and Diversification Findings

The correlation analysis in Section 6.1 found no negatively correlated pairs among the 30 stocks. So the diversification the portfolio achieved was a matter of degree, not direction. Sector groups like technology,

financials and energy showed clearly different levels of positive correlation with each other, but none of them consistently moved in opposite directions. That's a realistic feature of an equity-only portfolio, since all 30 holdings share exposure to broad market-wide forces like monetary policy, macroeconomic growth and investor sentiment, whatever sector they sit in. In practice, this means the diversification benefit seen in this study, a Sharpe Ratio improvement rather than a wholesale drop in market exposure, is a within-asset-class effect. Getting the stronger kind of diversification that comes from genuinely offsetting, negatively correlated positions would most likely mean going beyond U.S. public equities, into government bonds or other asset classes whose returns aren't so tied to equity market sentiment.

The allocation results in Section 6.5 back this up. The five largest holdings in the Maximum Sharpe Portfolio span Technology, Energy and Consumer sectors and together make up only 35.5% of total capital. That shows the optimisation process kept broad diversification even after reweighting for efficiency. It didn't get around the correlation structure from Section 6.1 by concentrating in a handful of assets. It got around it by shifting relative weight toward sectors that already offered meaningful diversification value.

7.4 Portfolio Behaviour During the COVID-19 Market Crash

The rolling volatility and drawdown results from Section 6.3 tell us more about how the portfolio handled acute stress. The spike in rolling volatility to 85.03% and the maximum drawdown of 33.24% both happened around the COVID-19 market shock of early 2020. That shows diversification across sectors, while it did reduce average portfolio risk over the full five years, couldn't protect the portfolio from a systemic, market-wide shock. This lines up with a well-known feature of financial markets under stress. Correlations between risk assets tend to rise sharply as investors pull back from risk broadly rather than selectively, which is sometimes described as diversification failing right when it's needed most. Because the correlation matrix in Section 6.1 was worked out over the full five-year sample, it blends the higher correlations of the crisis period with the more typical correlations of calmer periods. That means the diversification benefit suggested by the full-sample correlation matrix probably overstates how much protection was actually available during that period of acute stress.

Even so, the portfolio's recovery to new cumulative highs, visible in the drawdown series in Figure 6.5, shows the underlying holdings had enough long-term growth in them to more than make up for this episode within the five-year window studied. A sharp but temporary drawdown followed by a sustained recovery is broadly what you'd expect from diversified U.S. equity portfolios after the 2020 shock. It's also a reminder that the maximum drawdown reported in Section 6.3 reflects one peak-to-trough decline at a single point in time, not a permanent loss of capital.

7.5 Limitations of Modern Portfolio Theory in Practice

This study has several limitations that come built into the mean-variance framework it uses. The framework assumes that historical mean returns, variances and covariances are reliable stand-ins for their future values. The optimisation results in Section 6.4 use the same historical dataset both to work out the optimal allocation and to judge how well it performed, so the reported improvement in Sharpe Ratio carries in-sample bias and hasn't been tested on new, unseen data. Expected returns in particular are

known to be estimated with much more error than volatility or correlation, which means mean-variance optimisation can be thrown off by small changes in the return assumptions. That's a long-recognised weakness of the Markowitz approach used here.

The theory also assumes correlations between assets stay stable over time, an assumption that Section 7.4 directly challenges, since correlations across risk assets tend to rise during periods of crisis. On top of that, this analysis leaves out transaction costs, taxes and the practical work of rebalancing a portfolio back to its optimised weights over time, all of which would eat into the real-world advantage of the Maximum Sharpe Portfolio over the equal-weight benchmark. A portfolio that's optimal at one point in time will naturally drift away from that optimum as prices move, and rebalancing it back costs money. The risk-free rate was also held fixed at 4% for the whole study period, even though short-term interest rates moved around a fair bit between 2020 and 2025. So the Sharpe Ratios in this study are an approximation rather than a period-accurate risk-adjusted return.

7.6 Assumptions and Directions for Future Work

Several extensions would deal with the limitations above and would be natural next steps for this project. Adding transaction costs and a defined rebalancing schedule, quarterly rebalancing back to target weights for example, would give a more realistic sense of what risk-adjusted return is actually achievable rather than just theoretical. Out-of-sample or rolling-window testing, where the optimal weights are worked out on one period and tested on a later, unseen period, would give a tougher test of whether the Monte Carlo optimisation genuinely improves future performance or just fits historical noise. That would directly address the in-sample bias raised in Section 7.5.

The Black-Litterman model, which blends the market's implied equilibrium return with an investor's own views, offers a way to soften mean-variance optimisation's sensitivity to estimation error in expected returns, letting the analysis bring in some forward-looking judgement instead of relying purely on historical averages. Extending the analysis through factor investing, breaking returns down into exposures to established risk factors like size, value and momentum, would give a more detailed explanation of why certain holdings, including Apple, Exxon Mobil and Costco, contributed so much to the optimised portfolio's efficiency, and would help separate genuine diversification benefit from exposure to a handful of common underlying risk factors. Finally, extending the asset universe beyond U.S. public equities, to include government bonds for example, would test whether the stronger, direction-based diversification discussed in Section 7.3 is achievable in a way this equity-only universe doesn't allow.

7.7 Practical Implications for Investors

Put together, these findings point to a few practical takeaways for portfolio management. Periodic re-optimisation is worth doing on an ongoing basis rather than treating it as a one-off exercise, since even a systematic reweighting within the same 30-stock universe made a real difference to risk-adjusted return. Whether to lean toward a Maximum Sharpe or a Minimum Volatility orientation should follow directly from an investor's objective and time horizon, not a blanket preference for one over the other, since Section 7.2 shows the two portfolios serve genuinely different purposes. The largest optimised weights

landing on Apple, Exxon Mobil, Costco, Meta Platforms and Amazon, none of which was the single highest-returning stock in the sample, shows that a disciplined, correlation-aware allocation process can find efficient holdings that simple return-ranking would miss.

Finally, the gap between the portfolio's full-sample correlation structure and how it actually behaved during the COVID-19 shock, covered in Section 7.4, is a caution against leaning too heavily on long-run historical correlation and volatility estimates as a guide to diversification during periods of acute market stress. Investors and analysts using a similar approach should treat the resulting optimal weights as a reasonable long-run allocation guide, not a guarantee of protection during a systemic shock, and should think about pairing historical optimisation with stress-testing against past crisis periods.

8. Conclusion

This report has applied Modern Portfolio Theory to a diversified portfolio of 30 U.S. equities across seven sectors, working through data collection and descriptive analysis, correlation and covariance analysis, portfolio-level risk and return measurement, and Monte Carlo-based optimisation. That covers every research objective set out in Section 2.2.

The main finding is that the equal-weighted portfolio, while it delivered a solid expected annual return of 23.78% at an annualised volatility of 21.40%, wasn't the most efficient use of capital available. Reweighting toward Apple, Exxon Mobil, Costco, Meta Platforms and Amazon raised the Sharpe Ratio from 0.92 to 1.13 without a proportional rise in risk, while the Minimum Volatility Portfolio showed that cutting risk and improving risk-adjusted efficiency are two different goals that don't always move together. The correlation analysis also showed that the portfolio's diversification benefit came from combining assets across sectors with varying degrees of positive correlation, not from negatively correlated holdings, and that this diversification, while it reduced average risk, didn't stop a substantial, systemic drawdown during the COVID-19 market shock.

The main contribution of this project is a complete, reproducible portfolio analysis and optimisation pipeline, built from first principles in Python and checked against what Modern Portfolio Theory would predict, together with a straightforward account of where the theory's assumptions diverge from real markets. As a planned next step, the outputs from this project have been prepared for an interactive Streamlit dashboard, so the performance, risk and optimisation results here can be explored directly by a non-technical audience.

In short, this study shows that Modern Portfolio Theory works when applied to a real equity universe. A rigorous quantitative pipeline, covering descriptive statistics, covariance and correlation analysis, risk measurement and simulation-based optimisation, was used to show that disciplined capital allocation, not stock selection on its own, can deliver a meaningfully better risk-adjusted outcome. The resulting Maximum Sharpe and Minimum Volatility portfolios give investors a concrete, evidence-based choice between two clearly defined risk postures, and the project's honest treatment of its own limitations gives a solid foundation for extending it toward out-of-sample validation, factor-based and Black-Litterman

approaches, and diversification across more asset classes. Overall, this analysis stands as a solid demonstration of quantitative portfolio management, pairing technical rigour with clear, decision-relevant financial insight, and one that could be extended to more sophisticated investment mandates in future work.

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Appendix

A.1 Project Workflow

The workflow behind this analysis moved through the following stages:

- Download historical prices
- Data validation
- Return calculations
- Portfolio statistics
- Risk analysis
- Monte Carlo simulation
- Efficient Frontier
- Portfolio Analysis Report

A.2 Tools and Libraries

- yfinance: download of historical market data
- pandas: data manipulation and analysis
- NumPy: numerical computing and matrix operations
- Matplotlib: static data visualisation (Figures 6.2-6.6, 6.8)
- Seaborn: statistical visualisation, including the correlation heatmap (Figure 6.1)
- Streamlit: interactive dashboard for presenting the analysis

A.3 Summary of Key Metrics

Metric	Equal-Weight Portfolio	Maximum Sharpe Portfolio	Minimum Volatility Portfolio
Expected Annual Return	23.78%	29.53%	20.44%
Annualised Volatility	21.40%	22.60%	18.55%
Sharpe Ratio	0.92	1.13	0.89
Maximum Drawdown	33.24%	-	-
Average Rolling Volatility (30-day)	18.10%	-	-

A.4 Analysis Parameters

- Study period: 1 January 2020 to 1 January 2025 (five years of daily data).
- Annualisation convention: 252 trading days per year.
- Risk-free rate: 4% per annum, held constant throughout the study period.
- Monte Carlo simulation size: 10,000 randomly weighted portfolios, generated with a fixed random seed (`np.random.seed(18)`) to ensure reproducibility.
- Rolling volatility window: 30 trading days.

A.5 Author Information

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Project: Quant Foundations (Project 1)

This report was prepared from the author's original Jupyter Notebook analysis. All calculations, visualisations, figures, tables, and results presented herein were generated from the underlying analysis and represent the author's original work.